

ePortfolio Reflection Essay Draft

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Introduction

The capstone project I am working on involves Musical Bot Encore, a modular robot designed to play household keyboard instruments. Compared to older exhibit concepts like GuitarBot, Musical Bot Encore is intended to be a more compact and portable device that can mount non-invasively on a household keyboard, perform timed mechanical key presses to play selected digital music tracks, and demonstrate mechanical, electrical, and software engineering in a way that K-12 students and families visiting Engineering Expo can see and hear. During capstone, I have been involved in technical writing through the design proposal, self-directed learning in Fusion 360, CAD modeling and drawing support for the frame and solenoid layout, and collaboration through team planning documents such as the team charter.

Reviewing the NACE career readiness competencies made me realize that my capstone project was much more than a technical task. My capstone was essentially a workplace simulation that required me to demonstrate effective communication, critical thinking, technology, teamwork, leadership, professionalism, career and self-development, and equity and inclusion. All of these competencies apply to my goal of pursuing a career in aerospace or mechanical engineering, where I will have to combine design, analysis, modeling, communication, and professional discipline.

Communication

Effective communication is especially noticeable in engineering projects because the project team must communicate a large amount of information to different audiences at various stages. In particular, the design proposal required me to explain the Musical Bot Encore project idea to both MIME faculty and the project sponsor. Additionally, the scope and constraints had to be defined clearly so the design would also make sense for the target audience of future Engineering Expo visitors.

The robot is not intended to be a full-fledged music-producing device that could be used for professional recording. Instead, Musical Bot Encore is intended to play pre-selected songs in a reliable way, mount non-invasively on a household instrument, have a simple and quick assembly procedure, and provide visitors with a clear connection between a solenoid moving and its corresponding sound.

As a result, capstone helped me become familiar with the importance of communication as a key engineering task. It is rare that calculations alone can guarantee the successful completion of a

mechanical or aerospace project because engineers constantly have to explain their assumptions and tradeoffs to people who may not have the same technical background. Preparing written material and developing the design with CAD models and drawings helped me prepare for communicating similar messages to engineers, managers, technicians, and potential clients in the future.

Critical Thinking

The application of critical thinking to engineering projects is best explained by the connection between requirements, constraints, and design evidence. A specific example from my own work is related to the actuation mechanism. We initially assumed that a MOSFET driver board was available, but the tests we conducted showed that the hardware in question was actually a relay board. The relay board could not handle the fast switching speeds required to play certain musical passages, so the team was forced to reconsider the design instead of proceeding with implementation based on a false premise.

The design proposal mentioned the need for MOSFET switching, PWM control, and a step-by-step approach to testing, moving from a single key to a solenoid bank and finally to the full assembly. Critical thinking proved to be important during capstone because it required me to examine assumptions carefully and make informed choices about feasibility. Similarly, my future engineering career will rely on the ability to distinguish between assumptions and evidence, especially when design decisions affect reliability, cost, safety, and performance.

Technology

Capstone helped me become familiar with the importance of technology by introducing self-learning and its application in a real project. Since my initial Fusion 360 skills were somewhat limited, I decided to spend about seven hours of self-directed learning getting comfortable with the basics of the program. I did not just watch the Product Design Online tutorials; I completed each of them and built practice parts including a Lego handle, glass bottle, paper clip, and complex bottle models.

Afterward, I used the same technology to create Fusion 360 assembly models and drawings that could be used to evaluate the design of the robotic piano player. Specifically, the mechanical model of the frame helped us consider the location of the solenoid bank, determine the size of the assembly, and make sure there was sufficient clearance for attaching the system to the keyboard.

This experience helped me understand that self-learning can be a very helpful engineering tool as long as I can establish a connection between new knowledge and a relevant design challenge. As a result, capstone helped me prepare for future engineering careers where constant learning will be essential.

Specifically, learning new CAD, simulation, controls, Python automation, and aerospace design software will be important as I move into mechanical and aerospace engineering work.

Teamwork

Capstone helped me better understand the necessity of teamwork because there is no point in designing, building, and demonstrating an electromechanical robot entirely by myself. In particular, teamwork was evident in our use of the team charter, which gave the project team a clear set of expectations, roles, communication procedures, working norms, and accountability measures.

The need for cooperation became evident because our tasks were highly interdependent. When one person adjusted the mechanical design, the rest of the team had to consider the impact of those changes on electronics, the mounting structure, solenoid bank position, and setup procedure. Creating a mechanical design model in CAD also made it easier for my teammates to discuss design decisions with each other because the model gave us a shared visual reference.

I believe that this experience with teamwork will prove valuable in my future career because engineering projects are multidisciplinary in nature. Aerospace and mechanical engineering often involve collaboration between designers, analysts, controls specialists, manufacturing specialists, quality staff, and operations staff. In addition, capstone helped me learn the importance of documenting the design process so individual work can support the whole team.

Leadership

Leadership during capstone was not tied only to titles. Instead, it referred to the necessity of realizing which actions were important and taking them to ensure that the project kept moving in the right direction. For instance, as the project started facing difficulties related to design evaluation and visualization, I took initiative by improving my Fusion 360 skills and using those skills to create mechanical design models for the frame and solenoid bank.

Another example of leadership can be associated with maintaining the right design focus and preventing scope creep. While it is tempting to try to build an autonomous robot capable of reproducing any type of music, our project has a simpler and more realistic goal: we are supposed to build a reliable, portable, and visually appealing exhibit that demonstrates engineering principles in a way that is understandable to K-12 students and families visiting Engineering Expo.

All things considered, I will be able to apply the knowledge and skills acquired in capstone to take initiative and demonstrate leadership within future engineering teams. My previous experience in

wildland firefighting and EMT training taught me the importance of responsibility, calm communication, and dependable performance under pressure. The capstone experience added an engineering context to those lessons by showing that leadership also requires technical judgment, initiative, and follow-through.

Professionalism

Professionalism is an important aspect of capstone because it refers to the necessity of building and using reliable engineering solutions for real problems. Since Musical Bot Encore is expected to be demonstrated publicly during Engineering Expo, we have to make sure the robot is reliable, easy to assemble, relatively light, portable, and non-invasive to a household keyboard. For instance, some of the constraints include a student budget of about \$1,000, setup time of no more than ten minutes, maximum weight of no more than 100 pounds, and the ability to mount non-invasively and be easily disassembled. These factors require us to think about the reliability and practicality of the solution rather than just its appeal.

Apart from constraints, I have gained an understanding of how professionalism manifests itself in the form of well-written and well-organized documents. The team charter established clear expectations for roles, communication, work norms, and accountability. The ePortfolio also required clear filenames, organized folders, and linked PDF or DOCX files so that reviewers could understand the work without needing extra explanation. I believe that these standards will prove valuable in my future engineering career, where I will have to pay close attention to details and ensure professionalism in documentation.

Career & Self-Development

Compared to some other competencies, I felt particularly comfortable with career and self-development during capstone because I was able to see gaps in my skills and address them. Capstone required me to learn new tools and apply them to a real project. Specifically, while I knew that CAD modeling would be required for mechanical and aerospace engineering roles, I did not have enough practical Fusion 360 skill before capstone. As a result, I devoted about seven hours to self-learning Fusion 360.

Once I learned the basics, I created mechanical design models and drawings for the robotic piano player. This example shows that I can use capstone as a way to acquire new skills on an ongoing basis and apply them to engineering challenges. In the future, I plan to continue graduate study in mechanical engineering at Oregon State University before working in mechanical or aerospace engineering positions that will allow me to demonstrate design, analysis, modeling, and communication skills.

Equity & Inclusion

Capstone introduced me to the idea of equity and inclusion both in relation to teamwork and outreach activities. Since I am working on Musical Bot Encore, I can consider the target audience of the project to be Engineering Expo visitors, including K-12 students and their families. Because of that audience, it made sense to make the robot visually attractive and understandable to people who may not know much about robotics or engineering.

For instance, the robot is not a hidden system with only a screen showing the process. Instead, the design is intended to make the physical action visible so visitors can observe the cause-and-effect relationship between mechanical movement and the sounds made by the keyboard. As far as teamwork is concerned, the team charter helped set ground rules and norms for working together, which supports inclusion by making expectations clear for people with different working styles.

While this approach helped me understand the importance of inclusion as a design criterion, I can still gain more insight into how engineering decisions include or exclude different users. As a mechanical or aerospace engineer, I will regularly face the challenge of designing for diverse user groups and promoting equitable access to engineering solutions. Capstone helped me see inclusion as a design principle rather than just a separate social topic.

Conclusion

Overall, capstone was an interesting opportunity that helped me practice multiple NACE competencies in a real-world engineering environment. In particular, the Musical Bot Encore project required me to communicate, think critically, learn new technologies, coordinate with teammates, assume leadership responsibility, remain professional, develop personally, and promote inclusion. These competencies apply to my future plans related to obtaining a master's degree and getting a job in aerospace or mechanical engineering.

Specifically, I will be able to use capstone as a platform for building the right skills and habits that will eventually make me ready for a professional engineering career. The most valuable thing I have realized during capstone is that career readiness can be achieved through a set of engineering practices. Specifically, I improved at learning, analyzing, documenting, collaborating, and communicating.

Source Reviewed

National Association of Colleges and Employers. Career Readiness Competencies for a Career-Ready Workforce. Revised 2024.